Middlesex University London

Science and Technology

Computer Science

Computer Systems Architecture and Operating Systems

CST1500

Module Leader: Dr. Adam Philpot

2024/2025

Duration of the module: 12 Weeks

Document Version *1.1*

Online location of handbook

This handbook can also be accessed via My Learning at: https://mymdx.mdx.ac.uk/ via 'My Learning'.

Other formats available

This handbook is available in a large print format. If you would like a large print copy or have other requirements for the handbook, please contact the Disability Support Service disability@mdx.ac.uk

Disclaimer

The material in this handbook is as accurate as possible at the date of production. You will be notified of any minor changes promptly. If there are any major changes to the module you will be consulted prior to the changes being confirmed. Please check the version number on the front page of this handbook to ensure that you are using the most accurate information.

Other documents

Your module handbook should be read and used alongside your programme handbook and the information available to all students on My Learning and MyMDX including the Academic Regulations. Your programme handbook can be found on the My Learning programme page for your programme.

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1. Welcome

This module provides a basic understanding of computer architecture and the relationship between the hardware and software components of a computer system. It will enable students to utilise basic hardware, networking and operating systems theory and terminology to explain the structure of a computer and its operation.

2. The module teaching team

Please see below details of the teaching team for this module.

Module Leader: Dr. Adam Philpot		
	Room number:	TG03
	Email:	a.philpot@mdx.ac.uk
	Telephone number:	N/A
	Office hours:	Wednesday 14:00 – 16:00.
Lab Tutor: *Dr. Ali Khoshkholghi*		
	Room number:	T124
	Email:	a.khoshkholghi@mdx.ac.uk
	Telephone number:	N/A
	Office hours:	By appointment only.

3. Communication with the teaching team

Students may contact staff via e-mail (preferably), and by making an appointment to see them outside of classes during their office hours or arranged time.

The module leader and other module tutors are likely to send urgent and/or individual messages about the module to you by email, so it is important that you read your University email regularly.

If you have any queries, in the first instance, problems should be dealt with by talking to your tutor after the class.

You can give feedback on this module to the module leader, your Student Voice Leader, to your personal tutor, and through the end of module evaluation survey.

4. Module overview

Aims

The module aims to provide students with an understanding of the fundamental behaviour and components (hardware and software) of a typical computer system, and how they collaborate to manage resources and provide services in scales from small, embedded devices up to the global internet. The module will focus on the understanding of concepts, theory and associated terminology of two core components: 'Computer Architecture' and 'Operating Systems and Networks'. Practical hands-on laboratory experiments will be used to illustrate the application of theory and concepts. The lab work ranges between using C code, Batch, Bash and Python. This will allow students to develop a range of programming skills that will support their knowledge and skills not only in this module but in their future studies as well.

Learning Outcomes

On successful completion of this module, the student will be able to:

Knowledge

LO1	Demonstrate and explain the role and operation of the main hardware computer components (including processor, memory, and input/output components).
LO2	Demonstrate and explain how a computer's hardware and software work together to manage, process and transmit information.
LO3	Demonstrate and explain the role and variety of a range of mobile and fixed operating systems together with their features and functionalities.

Skills

LO4	Apply key theoretical and operational concepts and appropriate technical terminology to communicate information on the operation of computer hardware and operating systems (both verbally and by writing programs).
LO5	Conduct practical and theoretical research effectively in order to deepen understanding and appreciation of hardware and software components, and their relationship to solve given tasks easily.

LO6	Discuss likely future developments in the field of computer hardware, software, and advanced networks.

Syllabus

- Overview of computer systems, computer components, organisation, and architecture
- Data representations
- Number systems, characters, strings, image data, video data, audio data
- Introduction to digital and sequential logic
- Computer architecture
- Memory, Bus, ALU, Control Unit, Registers, CPU, Von Neumann Architecture, CISC, RISC, Multi-core processors
- Machine code and assembly Language
- Memory and storage systems
- Control Unit and microprogramming
- Introduction to Operating Systems
- System Structure and Processes
- Threads and Data Sharing
- Process Scheduling
- Process Synchronisation
- Deadlocks
- Memory Management
- Virtual Memory
- File Systems- I/O Systems
- Security Protection
- Network and Distributed systems

Learning and teaching strategy

Weekly Workshops/Seminars (2 hour). These sessions offer an inclusive environment for open discussions, concept explanations, group activities, and reinforcing understanding of security, theories, algorithms, and techniques. Through presentations, active student participation, and group discussions, the aim is to inspire students to explore deeper into topics of interest, fostering research-oriented engagement.

Weekly labs (3 hours). These sessions will be used to enable the students to further their knowledge, acquire and practise their research skills, debate the issues, to carry out practical activities and to work on individual technical projects. Each laboratory has a set number of exercises, which are also termed Students' Observation Behaviours (SOBs), where students are expected to perform practical tasks before advancing to the next session. Students can make use of the time given in these sessions to complete the SOB activities. During the lab session students will be offered help when needed especially in understanding the given SOBs and explaining ways to obtain the right output.

Weekly online student-focused workshops (1 hour). These sessions are student-led

and can host a range of activities listed above and provide amnesty for students who are falling behind in their SOBs and studies.

Briefly, the delivery of this module will be on-campus, over 12 weeks and have:
Weekly 2-hour workshops;
Weekly 3-hour labs;
Weekly drop-in session; and
Require additional 228 hours of independent study

Key concept videos and additional learning materials are provided on the 'Virtual Learning Environment' to help students with their study.

Furthermore, the module will run as follows;
Weeks 1 to 4 Computer Architecture;
Weeks 5 to 10 Operating Systems and Distributed Computing;
Week 11 to 12 Future Developments.

Assessment scheme

Formative assessment scheme:

An important element of overall learning strategy is to provide formative feedback on the work done in the laboratory session, and to provide a framework for guiding students on the work to be done during these sessions to improve the student performance. Tasks set will allow self-assessment moderated by tutors through a number of SOBs that will be required to be accomplished by the students during each lab session. SOBs are used as formative feedback and inform the assessments.

Three in class tests will be available for students to self-assess what they have learned from the lectures and the course at intermittent points over the 12 weeks. Test One will be in week 4, test two in week 8 and test three in week 11.

Summative assessment scheme:

There will be two pieces of coursework, each carry a weighting of 50%. Coursework one will be a portfolio of 10 SOBs to be submitted by the end of week 6. Coursework two will be a group project to be handed in by the end of week 12.

Scheduled teaching

Type	Hours
Independent study	228
Scheduled Teaching	72

Research Ethics

The teaching, learning, assessment and research activities undertaken in this module have been considered and are <u>not likely to require ethical approval</u>.

- However, please seek advice if undertaking the module entails carrying out any research activities involving **human participants, human data, animals/animal products, precious artefacts, materials or data systems.** If you submit work that includes data gathered from or about people, this may be treated as academic misconduct and could lead to fail grade being awarded.
- Further information about ethics can be found at the following link: http://mdx.mrooms.net/enrol/index.php?id=12277 (Log in required)

5. Learning resources

This module has a variety of learning resources available for you to use to support your learning. These include lecture videos, lecture slides, code examples, feedback, and key reading materials. These can be accessed online via the module page. Please visit the module page regularly to make use of these.

A reading list for this module can be found here:

KeyLinks

Additional resources are mentioned at relevant times in the weekly exercise documents.

6. Expectations of studying this module

The module team are here to help and support you achieve your goals. One of the key elements to successfully completing this module is engaging with all of the learning opportunities we offer as well and working with your peers to support one another.

This module is designed as a combination of contact sessions, directed study and independent study. This means you must participate in all the allocated sessions and you must complete all set prework and activities outside them. Students are expected to take an active part in all learning sessions whether these are online or on campus; lectures, lab sessions, practical classes, seminars and workshops.

To make the most of this module please complete the following every week

- Complete all prework in preparation for learning sessions. This may be watching videos, reading through set material or chapters and completing activities. Please make notes of points you need to clarify and discuss these in learning sessions with module tutors.
- Read through the notes making a note of any points you need to discuss with your tutor.
- Complete the set activities before the next session, making a note of any points you need to discuss with your tutor.
- Go to the module My Learning page, attempt the quizzes, make use of extra material, view the lecture videos, and access the activity examples. Make a note of anything you wish to discuss with your tutor.
- Complete further reading from the core text online.

The module team is committed to support you and your fellow students whilst you undertake this module. In order for you to get the most out of sessions you need to come prepared and ready to contribute. Please ensure that any work set by the team has been completed before workshops. After each class please review what has been covered and make a note of anything you would like clarification on.

Attendance and Engagement

Engaging with online and on-campus in-person learning and activities is integral to your success. Middlesex University supports you to achieve your full potential through a number of strategies, all of which provide a supportive learning environment online, remotely, face-to-face, or blended.

Further information on attendance and engaging with your programme will be available at your Induction and general information is available at the weblink below:

Home Page - MyMDX

Professional behaviour and online conduct

The programme of study you are undertaking is underpinned by developing professional behaviour and attitude. You are expected to behave in a professional, supportive manner to your peers and teachers. You must come to sessions prepared and ready to contribute where appropriate. Please remember that your University ID should be carried with you always whilst on campus and you must be able to identify yourself if asked to do so. Please conduct your email communication with fellow students, tutors and all relevant staff in a formal and courteous manner.

Academic Integrity and Misconduct

Academic Integrity is a set of principles and values to show that you work in a professional, honest and ethical way. You should be aware of the University's academic integrity and misconduct policies and procedures. Taking unfair advantage over other students in assessment is considered a serious offence by the University. Action will be taken against any student who contravenes the regulations through negligence, foolishness or deliberate intent. Academic misconduct takes several forms, in particular:

- **Plagiarism** – using extensive unacknowledged quotations from, or direct copying of, another person's work and presenting it for assessment as if it were your own effort. This includes the use of third party essay writing services.
- **Collusion** – working together with other students (without the tutor's permission), and presenting similar or identical work for assessment.
- **Infringement of Exam Room Rules** – Communication with another candidate, taking notes to your table in the exam room and/or referring to notes during the examination.
- **Self-Plagiarism** – including any material which is identical or substantially similar to material that has already been submitted by you for another assessment in the University or elsewhere.
- **Unauthorised use of Artificial Intelligence** – using artificial intelligence without referencing as such in your submission. AI is a powerful tool that can increase the productivity of programmers. In this module you are welcome to use AI tools like GitHub Copilot, ChatGPT and CodeWhisperer to help you to write code for your coursework. You are also welcome to use text generation tools like ChatGPT and Microsoft Copilot to help write your reports.

  AI tools can help you to program and write more efficiently, but they are not going to complete the coursework for you, they frequently make mistakes, and they are unlikely to follow the technology requirements in the coursework descriptions. If you use these tools to generate the code and report for you without understanding the technologies or coursework requirements, then you will get a low mark for this module.

  I strongly recommend that you spend time understanding how the programming languages (C, Bash, Batch, Python etc.) work and write code with these technologies without the help of AI. Then you will be in a much better position to use AI tools to help you to generate drafts of pieces of code for your project. Your understanding of the technologies will enable you to change this code to match the project requirements, integrate it, and fix bugs.

Links to the relevant University regulations and additional support resources can be found here:

Home Page - MyMDX

Student Success Essentials

The course includes useful information about how to approach your assessments and complete them with honesty. The course also describes what plagiarism (cheating) is and how to avoid it so you don't face any disciplinary action. For successfully completing this course, you will be awarded a certificate that will verify the knowledge you have gained. Certificates can be shared and promoted via LinkedIn and other digital channels. You will have to log into to MyMDX and then MyLearning to access the course. https://mdx.mrooms.net/course/view.php?id=17199

Full details on academic integrity and misconduct and the support available can be found at Home Page - MyMDX

The Academic Integrity and Misconduct policy is available in our Public Policy Statements (under Academic Quality) at: Policies | Middlesex University (mdx.ac.uk)

Referencing & Plagiarism: Suspected of plagiarism?:
http://libguides.mdx.ac.uk/c.php?g=322119&p=2155601

Referencing and avoiding plagiarism: Home Page - MyMDX

The Middlesex University Students' Union (MDXSU) Advice Service offers free and independent support in making an appeal, complaint or responding to any allegations of academic or non-academic misconduct.
https://www.mdxsu.com/advice

Extenuating circumstances

There may be difficult circumstances in your life that affect your ability to meet an assessment deadline or affect your performance in an assessment. These are known as extenuating circumstances or 'ECs'. Extenuating circumstances are exceptional, seriously adverse and outside of your control. For further information search 'Extenuating circumstances' in MyMDX or visit: Home Page - MyMDX

7. Assessment

7.1 Formative and Summative assessments

Formative assessment: Formative assessments help show that you are learning and understanding the material covered in this module and allow us to monitor your progress towards achieving the learning outcomes. Although formative assessments do not directly contribute to the overall module mark they do provide an important opportunity to receive feedback on your learning.

Formative assessment	Deadline
Test One	Week 4 – 14th October 2024
Test Two	Week 8 – 11th November 2024
Test Three	Week 11 – 2nd December 2024

Summative assessment: Summative assessment is used to check the level of learning for the module. It is summative because it is based on accumulated learning during the course. It is the summative assessment that determines the grade that you are awarded for the module.

The table below specifies the associated deadlines:

Summative assessment	Weighting	Deadline	Feedback
Coursework One	*50%*	*Week* 6 – 1st November 2024	*Up to 15 working days after the deadline*
Coursework Two	*50%*	*Week 12 – 13th December 2024*	*Up to 15 working days after the deadline*

In order to pass this module, you need to pass all assessments with a minimum grade of *40%* or equivalent.

In the event that Coursework One does not meet the minimum grade, ONE resubmission may be made before the end of the module – Week 12 - 13th December 2024. Students should submit the same assessment and draw on the feedback given in order to improve their grade. This second submission will be treated as a first attempt and **will not** be capped at 40%.

Before you submit your work for final grading, please ensure that you have accurately referenced the work. It is your responsibility to check the spelling and grammar, as all written assessments will assess technical proficiency in the English. This means accurate and effective spelling, punctuation and grammar. Details of

how it will be assessed will be provided in the marking criteria for each assessment and the University overall approach can be found within the Grade Criteria Guide in the University Regulations https://www.mdx.ac.uk/about-us/policies (scroll to university regulations)

Reasonable adjustments will be made for those students who have a declared disability/specific learning condition which would affect performance in this area.

Reassessment for this module normally takes place in the following way:

An uncapped resit coursework would be completed by students who have failed the module at the end of the semester. The resit coursework would be completed at the next available opportunity. More information can be found here: Home Page - MyMDX

Late Submissions

Late Submission means you can submit an individual written assignment up to 24 hours after the deadline without requesting an extension. However please note, the component grade will be reduced by the equivalent of 10% (except where this would reduce a pass grade below 40%). Students registered with the Disability and Dyslexia Service, who have an LSF that specifies additional requirements may be able to submit individual written assignments up to 5 days late without capping if this is indicated as a reasonable adjustment.

This option is available at the original deadline only for summative assessment. If you submit your work more than 24 hours after the original deadline, your work will not be marked and you will be given a grade of 20 (non-submission) for this piece of assessment.

Please note that late submission is not available for presentations, moots, vivas, performances, group work, SOBS, exams, in-class tests, portfolios, and some other assessments – please check with your module leader if you are not sure.

When selecting this option, you should take into consideration the impact this will have on your deadline schedule and your final grades, including your final degree classification.

7.2 Feedback on your assignments

You will be provided with feedback on all assessments that is helpful and informative, consistent with aiding the learning and development process. The nature of the feedback shall be determined at programme level but may take a variety of forms including: written comments; individual and group tutorial feedback; peer feedback; or

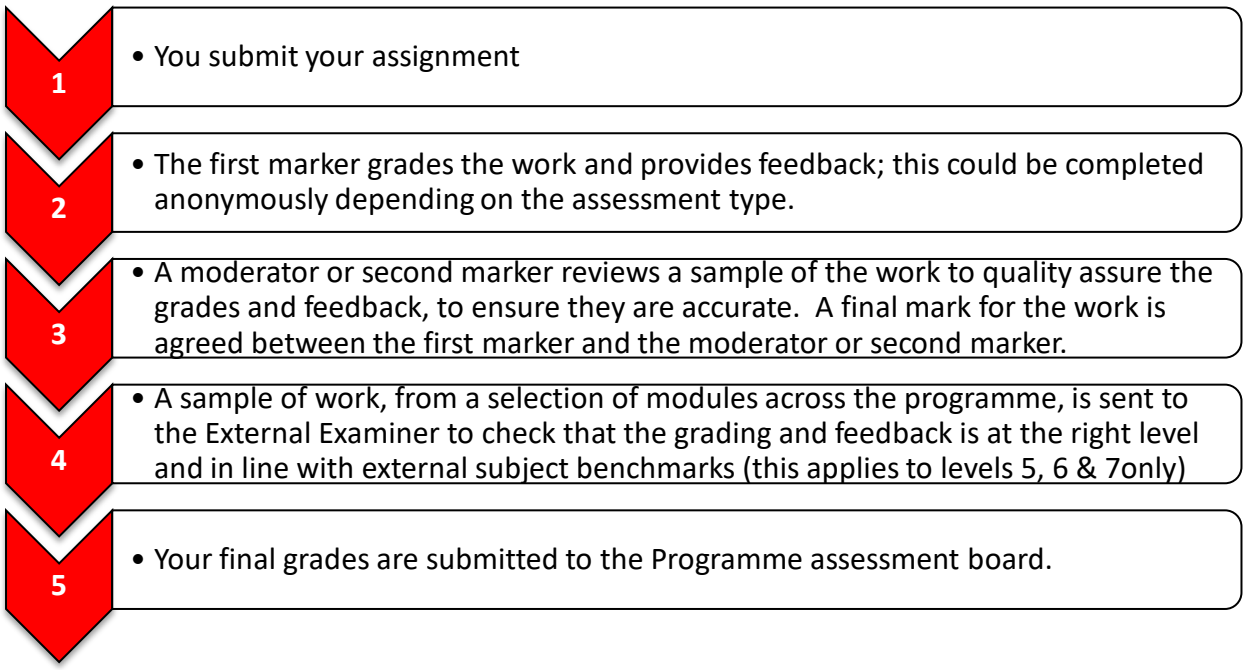
other forms of effective and efficient feedback. If you have submitted a formative or draft assessment, you will receive feedback but no grade. The comments should inform you about how well you have done or tell you about the areas for improvement. All assignments should be submitted online unless specified in assessment briefs.

Feedback on summative assessments will normally be provided within 15 WORKING DAYS of the published submission date.

7.3 How is your assignment mark agreed?

External Examiners (external academic experts) review what we deliver at a programme level. The University reviews a sample of your work to quality assure the grades and feedback you received from the person who marked your work. Our External Examiners will sample a selection of modules from a programme, with more focus of outcomes between modules within a programme.

The following diagram provides an overview of the marking process for your module assessment. Further information on the role of external examiners can be found at. https://www.mdx.ac.uk/about-us/policies/academic-quality/handbook (section 4)



Results Confirmation

First Semester

Provisional Grades: At the end of your first semester, you can see your module grades in the Grades and Progress tile within MyMDX. These grades are provisional and not yet confirmed.

Second Semester

Final Grades and Progression: After your second semester, the Programme Assessment Board will confirm your grades. Then, your final module results, progression status, or finalist classification will be posted in the Grades and Progress tile within MyMDX.

Need Help or More Info?

- **University Guide**: Find detailed information in the University Guide in the Grades and Progress tile within MyMDX.
- **Support Team**: Ask your Progression and Support Team Officer for advice.
- **Regulations**: Check the University regulations for more details.

7.4 Anonymous Marking Assessment Policy

We have worked with the Middlesex University Students' Union (MDXSU) to create an anonymous marking policy, in response to student feedback. Anonymous marking ensures that your identity (your name, student number and other personal/identifiable information) is not made available to academics when they are marking your work. This means that you can have confidence that your assessments will be marked fairly and consistently. However, there are some forms of assessment for which anonymity cannot be guaranteed and these are recognised in the policy. The Anonymous Marking Assessment Policy is available at: https://www.mdx.ac.uk/__data/assets/pdf_file/0037/563599/anonymous-marking-assessment-policy.pdf

We now look at each component of summative assessment for this module in detail. Each of the following tables provides an overview of the requirements. The support provided for each component along with the feedback arrangements, is also detailed below.

7.5.1 Coursework One – SOB Portfolio

Coursework One – SOB Portfolio	
Module code	CST1500
Module title	Computer Systems Architecture and Operating Systems
Submission date, time	Friday 1st November 2024 online submission link
Feedback type & date	Written feedback on submitted work via the original submission
Word count	No limit
Assignment type	SOB portfolio
Assignment structure, format and details	Full details to be found in the Coursework One specification document on the course page.
Assessed learning outcome (s)	LO 1 - 6
Assessment weighting %	50%
Key reading and learning resources	Lab exercise worksheets from Week 1 – 6 plus additional online resources provided.

The following table details the support you will be receiving for this assessment and the feedback opportunities you will have.

Support and draft feedback sessions for Coursework One – SOB Portfolio
Coursework briefing The Coursework One specification will be available on the course page. Dedicated time to discuss the coursework specification will be in the Week Two lab/workshop/online sessions.
Draft feedback opportunities Students can receive feedback on their coursework during the lab/workshop/online sessions as and when required.
Additional support SLAs will be in labs and workshops to assist with questions regarding the coursework.

7.5.2 Coursework Two – Python Groupwork

Coursework Two – Python Groupwork	
Module code	CST1500
Module title	Computer Systems Architecture and Operating Systems
Submission date, time	Friday 13th December 2024 online submission link
Feedback type & date	Written feedback on submitted work via the original submission
Word count	No limit
Assignment type	Software written using the Python programming language
Assignment structure, format and details	Full details to be found in the Coursework Two specification document on the course page.
Assessed learning outcome (s)	LO 4 – 6
Assessment weighting %	50%
Key reading and learning resources	Lab exercise worksheets from Week 7 – 10 plus additional online resources provided.

The following table details the support you will be receiving for this assessment and the feedback opportunities you will have.

Support and draft feedback sessions for Coursework One – SOB Portfolio
Coursework briefing The Coursework Two specification will be available on the course page. Dedicated time to discuss the coursework specification will be in the Week Eight lab/workshop/online sessions.
Draft feedback opportunities Students can receive feedback on their coursework during the lab/workshop/online sessions as and when required.
Additional support SLAs will be in labs and workshops to assist with questions regarding the coursework.

7.5.3 Formative Test One

Formative Test One	
Module code	CST1500
Module title	Computer Systems Architecture and Operating Systems
Test date	Week 4
Feedback type & date	Instant feedback after the test.
Description of test	Multiple choice test
Duration of test	30 minutes
Assessed learning outcome (s)	[LO 1-6]
Module weighting %	0%
Key reading and learning resources	Lecture slides 1 to 6 (inclusive)

7.5.4 Formative Test Two

Formative Test Two	
Module code	CST1500
Module title	Computer Systems Architecture and Operating Systems
Test date	Week 8
Feedback type & date	Instant feedback after the test.
Description of test	Multiple choices test
Duration of test	30 minutes
Assessed learning outcome (s)	[LO 1-6]
Module weighting %	0%
Key reading and learning resources	Lecture slides 7 to 13 (inclusive)

7.5.5 Formative Test Three

Formative Test Three	
Module code	CST1500
Module title	Computer Systems Architecture and Operating Systems
Test date	Week 11
Feedback type & date	Instant feedback after the test.
Description of exam	Multiple choices test
Duration of exam	30 minutes
Assessed learning outcome (s)	[LO 1-6]
Module weighting %	0%
Key reading and learning resources	Lecture slides 14 to 21 (inclusive)

8. Learning Planner

Week Beginning	Week	Lab	Student Activity	Summative Assessment	Formative Assessment
Sep 23rd	1	Introduction to module. Under the Lid. Introduction to the C programming language.	Weekly exercise sheet. Lectures 1 and 2.		
Sep 30th	2	Logical Operators using C code. C code decision making.	Weekly exercise sheet. Lectures 3, 4 and 5.		
Oct 7th	3	C code Standard I/O. C code using Loops.	Weekly exercise sheet. Lectures 6 and 7.		
Oct 14th	4	An Introduction to Batch Files and Scripting in Windows. Catch up.	Weekly exercise sheet. Lectures 8 and 9.		Test One
Oct 21st	5	Introduction to Unix Shell Scripting with Bash	Weekly exercise sheet. Lectures 10 and 11.		
Oct 28th	6	Coursework One support. Catch up. Reading week.	Weekly exercise sheet. Lectures 12 and 13.	SOB Portfolio Submission	

Nov 4th	7	Introduction to the Python programming language.	Weekly exercise sheet. Lectures 14 and 15.		
Nov 11th	8	Selection (if, then, else) in Python Programming.	Weekly exercise sheet. Lectures 16 and 17.		Test Two
Nov 18th	9	Python Multithreading. Synchronizing Threads.	Weekly exercise sheet. Lectures 18 and 19.		
Nov 25th	10	Python Catch up. Coursework Two group work.	Weekly exercise sheet. Lectures 20 and 21.		
Dec 2nd	11	Introduction to Arduino. Arduino basic examples.	Weekly exercise sheet. Lecture 22.		Test Three
Dec 9th	12	Arduino catch up. Coursework Two group work support.		Python Groupwork Submission	

9. University 20-point Scale

20-point scale	General scale	General scale (full ranges)	Percentage used for aggregation purposes only (for areas marking directly to the 20 point scale on modules with multiple assessment components)
1	80% - 100%	79.50% - 100%	90%
2	76% - 79%	75.50% - 79.49%	77.5%
3	73% - 75%	72.50% - 75.49%	74%
4	70% - 72%	69.50% - 72.49%	71%
5	67% - 69%	66.50% - 69.49%	68%
6	65% - 66%	64.50% - 66.49%	65.5%
7	62% - 64%	61.50% - 64.49%	63%
8	60% - 61%	59.50% - 61.49%	60.5%
9	57% - 59%	56.50% - 59.49%	58%
10	55% - 56%	54.50% - 56.49%	55.5%
11	52% - 54%	51.50% - 54.49%	53%
12	50% - 51%	49.50% - 51.49%	50.5%
13	47% - 49%	46.50% - 49.49%	48%
14	45% - 46%	44.50% - 46.49%	45.5%
15	42% - 44%	41.50% - 44.49%	43%
16	40% - 41%	39.50% - 41.49%	40.5%
17	35% - 39%	34.50% - 39.49%	37%
18	30% - 34%	29.50% - 34.49%	32%
19	0% - 29%	0.01% - 29.49%	15%
20	Non-participation	0%	0% (non-submission of a component)